

Lecture-12

- $X \sim \text{Bin}(n, p)$ $\rightarrow$ populaⁿ success probab.

$\hookrightarrow$ no. of successes in n indep. bernoulli trials

$$\hat{p} = \frac{X}{n}$$

$$E[X] = np$$

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{np}{n} = p$$

$\hat{p}$ is an unbiased estimator of p

$$\text{Var}(X) = np(1-p) \quad \text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{1}{n^2} np(1-p)$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

- for large n , CLT says $-\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx N(0,1)$

for $100(1-\alpha)\%$ CI

$$P\left(-Z_{\alpha/2} < \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} < Z_{\alpha/2}\right) \approx 1 - \alpha$$

$$p \approx \hat{p} \quad q \approx (1 - \hat{p})$$

We get Wald interval:

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\begin{cases} np \geq 5 \\ np(1-p) \geq 5 \end{cases}$$

Ex. $n=500$, $x=340$ subscribe, 95% CI

$$\hat{p} = \frac{340}{500} = 0.68$$

$$\alpha = 0.05, \alpha/2 = 0.025$$

$$Z_{0.025} = 1.96$$

$$0.68 \pm \underbrace{(1.96)(0.02)}_{0.0392}$$

$$\sqrt{\frac{0.68 \times 0.32}{500}}$$

$$0.68 \pm 0.0392 \Rightarrow (0.6408, 0.7192)$$

- the ordinary Wald int. behaves poorly for small n or when $p \sim 0/1$.

Wilson score interval

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

- then margin of error $e = z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$$\frac{e}{z_{\alpha/2}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \Rightarrow \frac{e^2}{z_{\alpha/2}^2} = \frac{\hat{p}(1-\hat{p})}{n}$$

$$n = \frac{\hat{p}(1-\hat{p}) z_{\alpha/2}^2}{e^2}$$

eg. CI = 95%, $\alpha = 0.05$, $\alpha/2 = 0.025$, $e = 0.02$

$$e = z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\left(\frac{0.02}{1.96} \right)^2 = \frac{(0.68)(0.32)}{n}$$

$$n = \frac{0.2176}{0.0001}$$

$$n = 2176$$

- But what if don't have $\hat{p}$, then no $(1-\hat{p})$. still we want that our sample size is large enough.

$$n = \frac{z_{\alpha/2}^2}{4e^2}$$

$$p(1-p) \leq \frac{1}{4}$$

this is the worst case variance for bernoulli var.

eg. 95% CI, $e = 0.02$

$$n = \frac{(1.96)^2}{4(0.02)^2} = \frac{3.8416}{0.0016} = 2401$$

- If we have 2 indep. proportions

$$X_1 \sim \text{Bin}(n_1, p_1), \quad X_2 \sim \text{Bin}(n_2, p_2)$$

$$\hat{p}_1 = \frac{X_1}{n_1}, \quad \hat{p}_2 = \frac{X_2}{n_2}, \quad \text{we want } \hat{p}_1 - \hat{p}_2$$

$$E[\hat{p}_1 - \hat{p}_2] = p_1 - p_2$$

$$\text{Var}(\hat{p}_1 - \hat{p}_2) = \frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}$$

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\text{Var}(\hat{p}_1 - \hat{p}_2)}$$

$$\text{large sample CI is } \left[(\hat{p}_1 - \hat{p}_2) \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \right]$$

$$\text{for hypothesis test of } H_0: p_1 = p_2, \quad \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

eg. $\hat{p}_1 = \frac{75}{1500}, \quad \hat{p}_2 = \frac{80}{2000}, \quad \alpha = 0.1, \quad \alpha/2 = 0.05$

$$\hat{p}_1 = 0.05, \quad \hat{p}_2 = 0.04$$

$$\hat{p}_1 - \hat{p}_2 = 0.01 \pm Z_{0.05} \sqrt{\frac{(0.05)(0.95)}{1500} + \frac{(0.04)(0.96)}{2000}}$$

$$\sqrt{0.0006 + 0.00048}$$

$$\sqrt{0.00108}$$

$$0.01 \pm \underbrace{(Z_{0.05})}_{1.645} (0.007) \Rightarrow 0.01 \pm 0.0115$$

$$\Rightarrow (-0.0015, 0.0215)$$

- $X_1, \dots, X_n \sim N(\mu, \sigma^2) \Rightarrow \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$P\left(\chi^2_{1-\alpha/2} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\alpha/2}\right) = 1-\alpha$$

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}} \right]$$

Eg. $n=10$ (46.4, 46.1, ...)
 95% CI for σ^2

$n-1=9$
 $\chi^2_{0.25,9} = 16.919$
 $\chi^2_{0.975,9} = 3.325$

$$\frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}}$$

$\bar{X} = \frac{461.2}{10} = 46.12$
 $S^2 = 0.2862$

$(n-1)S^2 = 9 \times 0.2862 = 2.5758$

$\Rightarrow \frac{2.5758}{19.023} < \sigma^2 < \frac{2.5758}{2.7}$
 $\Rightarrow (0.135, 0.954)$

if asked for σ just do $\sqrt{0.135}, \sqrt{0.954}$
 $\Rightarrow$ populaⁿ must be normal.

- 2 independent normal populaⁿ
 n_1, S_1^2 and n_2, S_2^2

$\frac{(n_1-1)S_1^2}{\sigma_1^2} \sim \chi^2_{n_1-1}$, $\frac{(n_2-1)S_2^2}{\sigma_2^2} \sim \chi^2_{n_2-1}$

$F = \frac{\chi^2_{v_1}/v_1}{\chi^2_{v_2}/v_2} \Rightarrow \boxed{\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1, n_2-1}}$

if $\theta = \frac{\sigma_1^2}{\sigma_2^2}$

$\boxed{\frac{S_1^2}{S_2^2} \cdot \frac{1}{F_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} \cdot F_{\alpha/2}(v_2, v_1)}$

Ex. $n_1 = 15$, $s_1 = 3.07$; $n_2 = 12$, $s_2 = 0.9$, 98% CI

$$s_1^2 = 9.4249, \quad s_2^2 = 0.81, \quad v_1 = 14, \quad v_2 = 11$$
$$\alpha = 0.02, \quad \alpha/2 = 0.01$$

$$F_{0.01}(14, 11) = 4.3, \quad F_{0.01}(11, 14) = 3.87$$

$$\frac{s_1^2}{s_2^2} = 11.64$$

$$\frac{(11.64)}{4.3} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{(11.64)(3.87)}{45.04}$$

$$2.7 < \sigma_1^2 / \sigma_2^2 < 45.04$$